

Cauchy - Euler Equation.

A diff. eq. of the form

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = \phi(x)$$

where $a_0, a_1, a_2, \dots, a_n$ are constants is called Cauchy - Euler diff. eq. This eq can be transformed into a linear diff. eq with constant coeffs by changing the independent variable x to z by substituting $x = e^z$ i.e. $z = \log x \therefore \frac{dz}{dx} = \frac{1}{x}$

$$\frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx} = \frac{1}{x} \frac{dy}{dz} \rightarrow (1)$$

$$\begin{aligned} \frac{d^2 y}{dx^2} &= \frac{d}{dx} \left[\frac{1}{x} \cdot \frac{dy}{dz} \right] \\ &= \frac{1}{x} \cdot \frac{d}{dx} \left(\frac{dy}{dz} \right) + \frac{dy}{dz} \frac{d}{dx} \left(\frac{1}{x} \right) \\ &= \frac{1}{x} \frac{d}{dz} \left(\frac{dy}{dz} \right) \cdot \frac{dz}{dx} + \frac{dy}{dz} \left(-\frac{1}{x^2} \right) \\ &= \frac{1}{x} \frac{d^2 y}{dz^2} \cdot \frac{1}{x} - \frac{1}{x^2} \frac{dy}{dz} \\ &= \frac{1}{x^2} \left(\frac{d^2 y}{dz^2} - \frac{dy}{dz} \right) \end{aligned} \rightarrow (2)$$

Writing D for $\frac{d}{dz}$ and D^2 for $\frac{d^2}{dz^2}$, above results may be written as

$$\frac{dy}{dx} = \frac{1}{x} D y \quad \text{i.e.} \quad x \frac{dy}{dx} = D y,$$

$$\frac{d^2 y}{dx^2} = \frac{1}{x^2} (D^2 - D) y \quad \text{i.e.} \quad x^2 \frac{d^2 y}{dx^2} = D(D-1) y$$

$$\text{Similarly} \quad x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2) y$$

$$\text{In general} \quad x^n \frac{d^n y}{dx^n} = D(D-1)(D-2) \dots (D-n+1) y$$

When above substitutions are made we get a

linear diff eq. with constant coeffs.

Solve (1) $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3$

(2) $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = \log x + x$

(3) $x^3 \frac{d^3y}{dx^3} - 3x^2 \frac{d^2y}{dx^2} + 6x \frac{dy}{dx} - 6y = 0$

(4) $x^3 \frac{d^3y}{dx^3} - x^2 \frac{d^2y}{dx^2} - 6x \frac{dy}{dx} + 18y = 0$

(1) $x^2 \frac{d^2y}{dx^2} - 2x \frac{dy}{dx} + 2y = x^3 \rightarrow (1)$

put $x = e^z$ i.e. $z = \log x$

$\therefore x \frac{dy}{dx} = Dy$, $x^2 \frac{d^2y}{dx^2} = D(D-1)y$ where $D = \frac{d}{dz}$

The transformed eq is

$$D(D-1)y - 2Dy + 2y = (e^z)^3$$

$$\text{i.e. } (D^2 - D - 2D + 2)y = e^{3z}$$

$$\text{i.e. } (D^2 - 3D + 2)y = e^{3z} \rightarrow (2)$$

This is a linear diff. eq with const coeffs

To find CF solve $(D^2 - 3D + 2)y = 0$

A.E. is $D^2 - 3D + 2 = 0$ i.e. $(D-1)(D-2) = 0$

$\therefore D = 1, 2$ CF = $c_1 e^z + c_2 e^{2z}$

PI = $\frac{1}{D^2 - 3D + 2} (e^{3z}) = \frac{1}{3^2 - 3(3) + 2} e^{3z} = \frac{1}{2} e^{3z}$

gen. soln of eq (2) is given by

$$y = c_1 e^z + c_2 e^{2z} + \frac{1}{2} e^{3z}$$

i.e. $y = c_1 x + c_2 x^2 + \frac{1}{2} x^3$

is the required soln

$$\left(\begin{array}{l} \because e^z = x \\ e^{2z} = (e^z)^2 = x^2 \\ e^{3z} = (e^z)^3 = x^3 \end{array} \right)$$

$$(2) \quad x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = \log x + x \longrightarrow (1)$$

Put $x = e^z$ then $z = \log x$

$$x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y, \quad \text{where } D \equiv \frac{d}{dz}$$

$\therefore$ The transformed eq is

$$D(D-1)y + Dy + y = z + e^z$$

$$\text{or } (D^2 - D + D + 1)y = z + e^z$$

$$\text{or } (D^2 + 1)y = z + e^z \longrightarrow (2)$$

To find CF, solve $(D^2 + 1)y = 0$

A.E. is given by $D^2 + 1 = 0$, $D = \pm i$

$$CF = C_1 \cos z + C_2 \sin z$$

$$PI = \frac{1}{D^2 + 1} (z + e^z) = \frac{1}{D^2 + 1} (z) + \frac{1}{D^2 + 1} (e^z)$$

$$= PI_1 + PI_2$$

$$PI_1 = \frac{1}{D^2 + 1} (z) = (1 + D^2)^{-1} z \\ = (1 - D^2 + D^4 - \dots) z \\ = z - 0 = z$$

$$PI_2 = \frac{1}{D^2 + 1} (e^z) = \frac{1}{2} e^z$$

$\therefore$ General soln of eq(2) is

$$y = C_1 \cos z + C_2 \sin z + z + \frac{1}{2} e^z$$

$$\text{or } y = C_1 \cos(\log x) + C_2 \sin(\log x) + \log x + \frac{x}{2}$$

$$(3) \quad x^3 \frac{d^3 y}{dx^3} - 3x^2 \frac{d^2 y}{dx^2} + 6x \frac{dy}{dx} - 6y = 0 \longrightarrow (1)$$

Put $x = e^z$, then $z = \log x$

$$x \frac{dy}{dx} = Dy, \quad x^2 \frac{d^2 y}{dx^2} = D(D-1)y$$

$$x^3 \frac{d^3 y}{dx^3} = D(D-1)(D-2)y \quad \text{where } D = \frac{d}{dz}$$

The transformed eq is

$$D(D-1)(D-2)y - 3D(D-1)y + 6Dy - 6y = 0$$

$$\text{i.e. } (D(D^2 - 3D + 2) - 3D(D-1) + 6D - 6)y = 0$$

$$\text{i.e. } (D^3 - 3D^2 + 2D - 3D^2 + 3D + 6D - 6)y = 0$$

$$\text{i.e. } (D^3 - 6D^2 + 11D - 6)y = 0 \quad \longrightarrow (2)$$

A.E. is given by $D^3 - 6D^2 + 11D - 6 = 0$

$D=1$ is a solution of this cubic equation

Dividing $D^3 - 6D^2 + 11D - 6$ by $D-1$,

$$D^3 - 6D^2 + 11D - 6 = (D-1)(D^2 - 5D + 6)$$

$$\therefore (D-1)(D-2)(D-3) = 0$$

$$D = 1, 2, 3$$

Soln of eq (2) is given by

$$y = C_1 e^x + C_2 e^{2x} + C_3 e^{3x}$$

i.e. $y = C_1 x + C_2 x^2 + C_3 x^3$ is the required soln.

Solve (1) $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} - 4y = x$

(2) $x^2 \frac{d^2 y}{dx^2} + 2x \frac{dy}{dx} - 20y = x^4$

(3) $x^3 \frac{d^3 y}{dx^3} - x^2 \frac{d^2 y}{dx^2} - 6x \frac{dy}{dx} + 18y = 0$

$$\begin{array}{r} D-1 \) \ D^3 - 6D^2 + 11D - 6 \ (D^2 - 5D + 6) \\ \underline{D^3 - D^2} \\ -5D^2 + 11D \\ \underline{-5D^2 + 5D} \\ 6D - 6 \\ \underline{6D - 6} \\ 0 \end{array}$$

Simultaneous Differential Equations

consider two diff. eqs of the form

$$\frac{dx}{dt} + a_1x + a_2y = f_1(t)$$

$$\frac{dy}{dt} + b_1x + b_2y = f_2(t)$$

Put $D \equiv \frac{d}{dt}$

$\therefore$ we get $Dx + a_1x + a_2y = f_1(t)$
 $Dy + b_1x + b_2y = f_2(t)$

i. $(D + a_1)x + a_2y = f_1(t) \rightarrow (1)$

$b_1x + (D + b_2)y = f_2(t) \rightarrow (2)$

Eliminate one of the variables x (or y), then we get a second diff. eq in y and t (or x and t). Solving this diff. eq we can find y (or x) in terms of t . Sub in (1) or (2) we can find x (or y) in terms of t .

Solve (i) $\frac{dx}{dt} - y = t, \quad \frac{dy}{dt} + x = 1$

(ii) $\frac{dx}{dt} + x = y + e^t, \quad \frac{dy}{dt} + y = x + e^t$

(i) $Dx - y = t \rightarrow (1)$

$x + Dy = 1 \rightarrow (2)$

(1) $\times D, \quad D^2x - Dy = D(t) \Rightarrow D^2x - Dy = 1 \rightarrow (3)$

(3) $+ (2), \quad D^2x + x = 2 \Rightarrow (D^2 + 1)x = 2 \rightarrow (4)$

A.E is given by $D^2 + 1 = 0, \quad D = \pm i$

CF = $C_1 \cos t + C_2 \sin t$

PI = $\frac{1}{D^2 + 1} (2) = \frac{1}{D^2 + 1} (2e^{0x}) = \frac{1}{0+1} (2) = 2$

Gen soln of (4),

$x = C_1 \cos t + C_2 \sin t + 2$

From eq (1)

$$y = Dx - t$$

$$= D[C_1 \cos t + C_2 \sin t + 2] - t$$

$$= -C_1 \sin t + C_2 \cos t - t$$

Required solution is given by

$$x = C_1 \cos t + C_2 \sin t + 2$$

$$y = -C_1 \sin t + C_2 \cos t - t$$

$$(ii) \quad \frac{dx}{dt} + x = y + e^t ; \quad \frac{dy}{dt} + y = x + e^t$$

$$\text{or } Dx + x - y = e^t, \quad Dy + y - x = e^t$$

$$\text{or } (D+1)x - y = e^t \rightarrow (1)$$

$$-x + (D+1)y = e^t \rightarrow (2)$$

$$(1) \times (D+1), (D+1)^2 x - (D+1)y = (D+1)e^t \dots$$

$$\text{or } (D+1)^2 x - (D+1)y = 2e^t \rightarrow (3)$$

$$(2) + (3), (D+1)^2 x - x = 3e^t$$

$$(D^2 + 2D + 1 - 1)x = 3e^t$$

$$\text{or } (D^2 + 2D)x = 3e^t \rightarrow (4)$$

$$\text{A.E. is } D^2 + 2D = 0, \quad D(D+2) = 0 \Rightarrow D = 0, -2$$

$$\text{CF} = C_1 e^{0t} + C_2 e^{-2t} = C_1 + C_2 e^{-2t}$$

$$\text{PI} = \frac{1}{D^2 + 2D} (3e^t) = \frac{1}{3} (3e^t) = e^t$$

$\therefore$ solution of eq (4) is given by

$$x = C_1 + C_2 e^{-2t} + e^t$$

$$y = (D+1)x - e^t$$

$$= (D+1)(C_1 + C_2 e^{-2t} + e^t) - e^t$$

$$= -2C_2 e^{-2t} + e^t + C_1 + C_2 e^{-2t} + e^t - e^t$$

$$y = C_1 - C_2 e^{-2t} + e^t$$

Required soln is given by

$$x = c_1 + c_2 e^{-2t} + e^t$$

$$y = c_1 - c_2 e^{-2t} + e^t$$

Solve (1) $\frac{dx}{dt} - 7x + y = 0$, $\frac{dy}{dt} - 2x - 5y = 0$

$$(D-7)x + y = 0, \quad (D-5)y - 2x = 0$$

$$(D-7)x + y = 0 \rightarrow (1)$$

$$-2x + (D-5)y = 0 \rightarrow (2)$$

$$(1) \times (D-5) \quad (D-5)(D-7)x + (D-5)y = 0 \rightarrow (3)$$

$$-2x + (D-5)y = 0 \rightarrow (2)$$

$$(3) - (2), \quad (D-5)(D-7)x + 2x = 0$$

$$(D^2 - 12D + 35 + 2)x = 0$$

$$\therefore (D^2 - 12D + 37)x = 0$$

$$\text{A.E.} \quad D^2 - 12D + 37 = 0$$

$$D = \frac{12 \pm \sqrt{144 - 148}}{2} = \frac{12 \pm \sqrt{-4}}{2}$$

$$= \frac{12 \pm 2i}{2} = 6 \pm i$$

$$x = e^{6t} [c_1 \cos t + c_2 \sin t]$$

From eq (1) $y = -(D-7)x$
 $= -(D-7) [e^{6t} (c_1 \cos t + c_2 \sin t)]$

$$\therefore y = - (e^{6t} (-c_1 \sin t + c_2 \cos t) + 7e^{6t} (c_1 \cos t + c_2 \sin t))$$

$$= e^{6t} [c_1 \sin t - c_2 \cos t + c_1 \cos t + c_2 \sin t]$$

$$= e^{6t} [(c_1 - c_2) \cos t + (c_1 + c_2) \sin t]$$

Solve (1) $\frac{dx}{dt} + 4x + 3y = t$, $\frac{dy}{dt} + 2x + 5y = e^t$

(2) $\frac{dx}{dt} - y = t$, $\frac{dy}{dt} + x = t^2$